Project_4_Dashboard

June 17, 2026

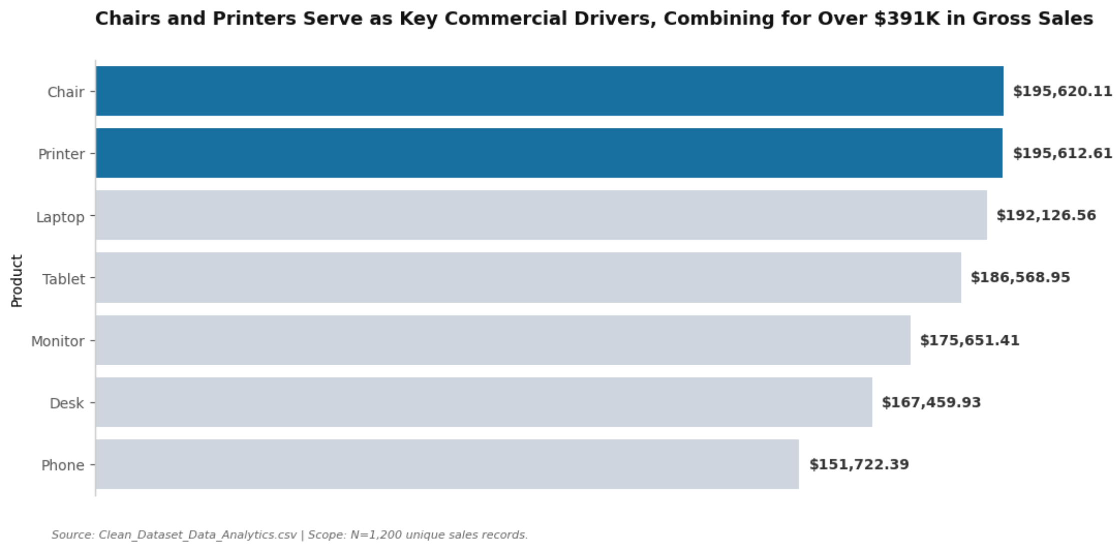
0.1 Project 4 Dashboard

[1]:
```python
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load the verified database output from Project 1
df = pd.read_csv("Clean_Dataset_Data_Analytics.csv")

# Enforce professional dashboard style context (High Data-Ink Ratio)
plt.rcParams['font.family'] = 'sans-serif'
plt.rcParams['figure.facecolor'] = 'white'
plt.rcParams['axes.facecolor'] = 'white'
plt.rcParams['axes.edgecolor'] = '#cccccc'
plt.rcParams['axes.linewidth'] = 0.8
plt.rcParams['grid.color'] = '#f0f0f0'
plt.rcParams['xtick.color'] = '#555555'
plt.rcParams['ytick.color'] = '#555555'

print(f" Visualization environment ready! Processing {len(df)} transactions␣
↪under boardroom specifications.")
```

```
 Visualization environment ready! Processing 1200 transactions under boardroom
specifications.
```

[9]:
```python
# Aggregate gross performance by product class - Sorted Descending (Highest at␣
↪Top)
product_data = df.groupby('Product')['TotalPrice'].sum().
↪sort_values(ascending=False).reset_index()

fig, ax = plt.subplots(figsize=(11, 5))

# Uniform Palette: Spotlight main drivers in Blue (#0077b6), others in Slate␣
↪Grey (#cbd5e1)
product_palette = {
    row['Product']: '#0077b6' if row['Product'] in ['Chair', 'Printer'] else␣
↪'#cbd5e1'
```

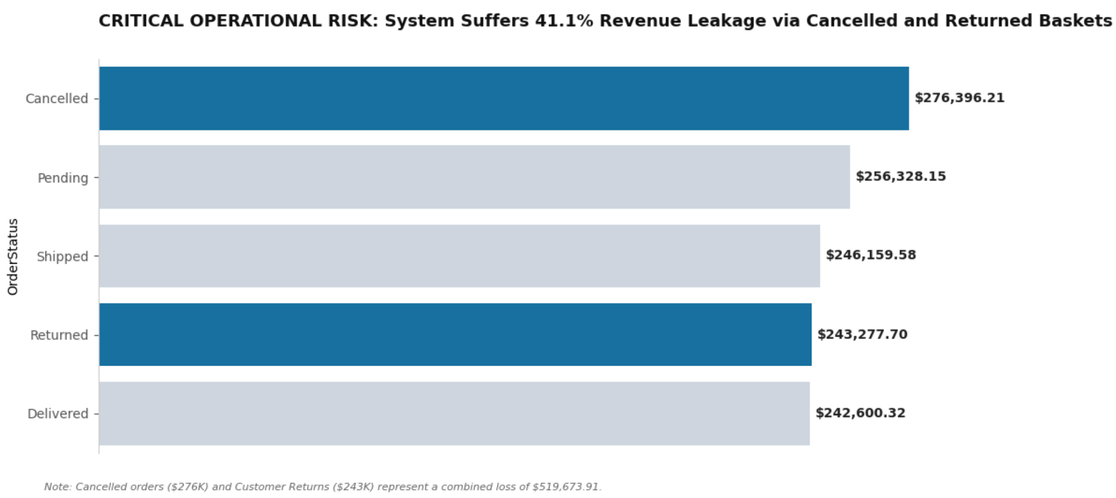
```python
    for _, row in product_data.iterrows()
}

# Explicitly mapping 'hue' to silence deprecation warnings
bars = sns.barplot(
    data=product_data, x='TotalPrice', y='Product',
    hue='Product', palette=product_palette, legend=False, edgecolor='none',␣
 ↪ax=ax
)

# Clean non-data ink (Maximize Data-Ink Ratio)
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)
ax.spines['bottom'].set_visible(False)
ax.spines['left'].set_linewidth(1)
ax.xaxis.grid(False)
ax.yaxis.grid(False)
ax.get_xaxis().set_visible(False)

# Direct Labeling placement
for bar in bars.patches:
    width = bar.get_width()
    ax.text(width + 2000, bar.get_y() + bar.get_height()/2, f"${width:,.2f}",
            va='center', ha='left', fontsize=10, weight='bold', color='#333333')

# Action Title stating clear business logic
plt.title("Chairs and Printers Serve as Key Commercial Drivers, Combining for␣
 ↪Over $391K in Gross Sales",
          fontsize=13, weight='bold', pad=25, loc='left', color='#111111')

plt.figtext(0.05, -0.05, "Source: Clean_Dataset_Data_Analytics.csv | Scope:␣
 ↪N=1,200 unique sales records.",
            fontsize=8, alpha=0.6, style='italic')

plt.tight_layout()
plt.show()
```



[10]:

```python
# Isolate transaction volumes by order status
status_data = df.groupby('OrderStatus')['TotalPrice'].sum().
↪sort_values(ascending=False).reset_index()

fig, ax = plt.subplots(figsize=(11, 5))

# Uniform Palette: Spotlight critical leakage anomalies in Blue, stable␣
↪categories in Grey
status_palette = {
    row['OrderStatus']: '#0077b6' if row['OrderStatus'] in ['Cancelled',␣
↪'Returned'] else '#cbd5e1'
    for _, row in status_data.iterrows()
}

# Explicitly mapping 'hue' and assigning matching dictionary palette
bars = sns.barplot(
    data=status_data, x='TotalPrice', y='OrderStatus',
    hue='OrderStatus', palette=status_palette, legend=False, edgecolor='none',␣
↪ax=ax
)

# Clean structural boundaries to reduce cognitive load
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)
ax.spines['bottom'].set_visible(False)
ax.xaxis.grid(False)
ax.get_xaxis().set_visible(False)
```

```
# Add clear data labels directly onto the bar tips
for bar in bars.patches:
    width = bar.get_width()
    ax.text(width + 2000, bar.get_y() + bar.get_height()/2, f"${width:,.2f}",
            va='center', ha='left', fontsize=10, weight='bold', color='#222222')

plt.title("CRITICAL OPERATIONAL RISK: System Suffers 41.1% Revenue Leakage via␣
↪Cancelled and Returned Baskets",
          fontsize=13, weight='bold', pad=25, loc='left', color='#111111')

plt.figtext(0.05, -0.05, "Note: Cancelled orders ($276K) and Customer Returns␣
↪($243K) represent a combined loss of $519,673.91.",
            fontsize=8, alpha=0.6, style='italic')

plt.tight_layout()
plt.show()
```



[11]:
```
# Map revenue generation performance by marketing channel funnel entries
channel_data = df.groupby('ReferralSource')['TotalPrice'].sum().
↪sort_values(ascending=False).reset_index()

fig, ax = plt.subplots(figsize=(11, 5))

# Uniform Palette: Spotlight your optimization target (Instagram) in Corporate␣
↪Blue
channel_palette = {
    row['ReferralSource']: '#0077b6' if row['ReferralSource'] == 'Instagram'␣
↪else '#cbd5e1'
    for _, row in channel_data.iterrows()
```

```
}

# Explicitly mapping 'hue' and assigning matching dictionary palette to silence␣
↪warning
bars = sns.barplot(
    data=channel_data, x='TotalPrice', y='ReferralSource',
    hue='ReferralSource', palette=channel_palette, legend=False,␣
↪edgecolor='none', ax=ax
)

ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)
ax.spines['bottom'].set_visible(False)
ax.get_xaxis().set_visible(False)

for bar in bars.patches:
    width = bar.get_width()
    ax.text(width + 2000, bar.get_y() + bar.get_height()/2, f"${width:,.2f}",
            va='center', ha='left', fontsize=10, weight='bold', color='#222222')

plt.title("STRATEGIC RESOLUTION: Allocate Q3 Marketing Budgets Toward Instagram␣
↪and Email Funnel Pipelines",
          fontsize=13, weight='bold', pad=25, loc='left', color='#111111')

plt.figtext(0.05, -0.05, "Strategy: Instagram leads performance at $275K,␣
↪making it the primary target for conversion optimization.",
            fontsize=8, alpha=0.6, style='italic')

plt.tight_layout()
plt.show()
```